VIGYAT SINGH

AI & Machine Learning Engineer | Final Year CSE Undergrad

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Portfolio | LinkedIn | GitHub

EDUCATION

KIIT University, Bhubaneswar **Aug 2022 – May 2026**

Bachelor of Technology in Computer Science and Engineering **CGPA: 8.44/10.0**

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Operating Systems, Computer Networks

EXPERIENCE

SDE-I **Feb 2026 – Present**

Quadrical AI, Gurugram, India (On-site)

- Engineered a **KPI calculation engine** for a solar energy monitoring platform using **FastAPI**, **Polars**, and **SQLAlchemy async**, computing 10+ KPIs across multi-plant portfolios via a **single optimised database fetch** with multi-dimensional grouping (all, plant, date, plant-date) returning all views in one API response.
- Architected the backend for a **CMMS** from scratch using **GraphQL (Strawberry)**, building ticket management, status workflow engine (Open → In Progress → Resolved → Closed), preventive vs. corrective maintenance classification, and inventory integration for spare parts tracking across solar plant sites.
- Built **RESTful widget endpoints** (portfolio status, card, earth map, table, plant status) with **Redis** caching (TTL 300s), serving aggregated fleet data across single-plant and multi-plant views.

AI Engineer (Contract) **Sep 2025 – Nov 2025**

Thinkspan (AI Agent Orchestration Platform), London, UK (Remote)

- Engineered **LLM-driven agent workflows** implementing data-driven task decomposition and output evaluation to ensure reliability in multi-agent systems.
- Optimized **orchestration logic** through rigorous experimentation, directly driving improvements in workflow efficiency and platform usability.

Data Science Intern **Jan 2025 – Jun 2025**

EdFoal Services Pvt. Ltd., Noida, India

- Engineered Python-based ETL pipelines processing **100K+ records**, achieving **20% improvement** in throughput through optimized data validation logic and parallel processing techniques.
- Built automated dashboard reporting system reducing manual effort from **2 hours to 45 minutes** (62.5% reduction), enabling real-time business intelligence.

TECHNICAL SKILLS

Languages: Python, Java, SQL

ML/AI: TensorFlow, PyTorch, Scikit-learn, YOLOv8, LangChain, FAISS, RAG Systems, Prompt Engineering

Data Engineering: Pandas, Polars, NumPy, PostgreSQL, MongoDB, ETL Pipelines, SupaBase

Backend: FastAPI, Flask, REST APIs, GraphQL, Strawberry, SQLAlchemy, Redis

DevOps & Tools: Docker, Git, CI/CD, N8N Automation

TECHNICAL PROJECTS

SaaS Opportunity Discovery Engine | N8N, Groq, Telegram, OpenRouter *Jan 2026*

- Architected an end-to-end **N8N workflow** scraping **Reddit, HackerNews, and ProductHunt** to capture real-time SaaS opportunities, achieving a processing rate of **25 leads daily**.
- Engineered an intelligent lead scoring system using **Groq LLaMA 3.3 70B** with automated **Telegram delivery** for instant insights.

AI-Powered Lead Intelligence Platform | GitHub • Medium *Python, Streamlit, LLMs, Google Sheets*

- Architected LLM-driven pipeline for Reddit lead qualification and outreach automation achieving **90%+ precision** with dynamic model fallback (LLaMA-3.3-70B → LLaMA-3.1-8B) ensuring **99% uptime**.

MCP-Integrated Agentic AI Chatbot | GitHub • Live *FastAPI, React, TypeScript, Supabase*

- Architected production-grade agentic AI system with **MCP** integration implementing 7 real-time tools with intelligent tool selection, dual-layer storage achieving **5ms context retrieval**.

Multi-Agent RAG System | Live • GitHub *FastAPI, FAISS, React, TypeScript*

- Built scalable RAG system with vector embeddings reducing query response to **2 seconds** at **95%+ accuracy**, minimizing hallucinations by **80%** via multi-stage verification.

PUBLICATIONS

IEEE IC-ICNS 2026 — "A Hybrid and Adaptive Deep Learning Framework for Robust Image Classification Across Diverse Domains" (Singh, V., Singh, P., & Bhuyan, G.) — Working Paper

Medium Technical Blog — "Building an AI Sales Intern That Works 24/7" — Article on designing a local AI-based lead-generation system using Reddit data, LLMs, real-time data ingestion, and intelligent fallback logic.

HONORS & LEADERSHIP

Competition Awards: Poster Winner at IIT Kharagpur (2024) · 2nd Place Launchpad Quiz (KIIT, 2024)

Hackathons: IIT KGP Data Science Hackathon (2026) — **Top 10 Teams** · Amazon ML Challenge 2025 Participant

Technical Recognition: HackerRank 5⋆ Python (Gold Badge) · Yuva Shikriti Award (2020)

Leadership: Founding Member, Writing Wing at KIIT Kounsel Society · Head of Media & Entertainment Wing, Quorient

Creative & Community: Independent Music Artist on Spotify · Open-source contributor